Lab Assignment: Building a Full-Stack E-Commerce Application

Part 1 — Product Listing Website

Objective

Design and develop a functional e-commerce website in three stages. The application will evolve from a basic product listing website into an application with shopping cart functionality, authentication, authorization, database migrations, logging, and monitoring.

This exercise is intended to introduce you to the architecture and engineering practices involved in developing a full-stack application.

Part 1: Product Listing Website

Build a basic e-commerce website that allows users to browse available products.

Requirements

1. Frontend

- Display a list of products.
- Show the name, description, price, and image of each product.
- Provide a product details page or view.
- Include a basic search or filtering facility.

2. Backend

- Implement a backend using a web framework of your choice.
- Expose API endpoints to retrieve the product catalogue and individual product details.
- Validate incoming requests and return appropriate HTTP status codes.

3. SQL Database

- Use a relational database to store product information.
- Design an appropriate schema.
- Populate the database with sample products.
- Ensure that product information is retrieved through the backend rather than accessed directly by the frontend.

4. Integration

- Ensure that the frontend, backend, and database work together as a single application.
- Provide instructions for running the application on a single computer.

Expected Outcome

A working product listing website backed by a relational database.

Submission Requirements — Part 1

- Submit a Git repository containing the frontend and backend source code.

- Include the database schema or setup scripts and the sample product data needed to initialize the database.
- Include a README with prerequisites, configuration, setup instructions, and commands to run the application locally.
- Include a brief description of the application architecture and the API endpoints implemented.
- Include screenshots of the product listing page and product details view.
- Ensure that another person can run the application on a single computer by following the README.